# Brevan Ellefsen

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## Research Interests

3-manifolds; measured laminations; essential surfaces and foliations; normal surface theory; pseudo-Anosov flows and dynamics; knot theory.

## Education

**University of Illinois Urbana-Champaign** — Champaign, Illinois
Ph.D. in Mathematics, expected May 2027; advisor: Nathan Dunfield — 2021–present

**Brigham Young University** — Provo, Utah
B.S. in Mathematics — 2020

## Publications and Preprints

1. **Brevan Ellefsen**. *Growth Rate of Essential Surfaces via Measured Laminations*. 2026. arXiv:2608.08833 [math.GT].

## Research in Progress

- **Least-Weight Laminations.** Developing a normal-surface theory of least-weight representatives of essential measured laminations.
- **Volume Bounds on Growth.** Investigating bounds on measured-lamination-space dimension, and hence essential-surface growth, in terms of hyperbolic volume.
- **Minimal Laminations.** Studying when essential laminations admit representatives whose leaves are minimal for a suitable ambient metric.

## Invited Research Talks

- **Shanghai Institute for Mathematics and Interdisciplinary Sciences (SIMIS)** — Online (two-part invited seminar), October 2026 (upcoming)
  *Growth Rate of Essential Surfaces via Measured Laminations*
- **University of Wisconsin–Madison** — Madison, Wisconsin, September 16, 2026 (upcoming)
  *Growth Rate of Essential Surfaces via Measured Laminations*
- **AMS Fall Central Sectional Meeting – Special Session on Foliations and 3-Manifolds** — St. Louis, Missouri, October 18–19, 2025
  *Growth Rate of Essential Surfaces via Measured Laminations*

## Contributed Research Talks

- **Topological and Geometric Structures in Low Dimensions** — SLMath, March 2026
  *Growth Rate of Essential Surfaces via Measured Laminations* (lightning talk)
- **Topological and Geometric Structures in Low Dimensions (Summer Workshop)** — SLMath, July–August 2025
  *Growth Rate of Essential Surfaces via Measured Laminations* (lightning talk)

- **42nd Workshop in Geometric Topology** — Oberlin College
  *Growth Rate of Essential Surfaces via Measured Laminations* — June 12–14, 2025
- **Log Cabin Workshop on Foliations** — St. George, Utah
  *Universal Circles, Part 3* (expository talk) — January 13–17, 2025

## Selected Research Programs and Conferences

**Topological and Geometric Structures in Low Dimensions** — SLMath
Program Associate — March 16–April 24, 2026

**Computational Problems about 3-Manifolds, Associated Groups and Varieties** — Rutgers University
Participant — October 4–5, 2025

**The St. Louis Topology Conference: Flows and Foliations in 3-Manifolds** — St. Louis, Missouri
Participant — May 17–19, 2024

## Honors and Funding

**Departmental Fellowship, University of Illinois Urbana-Champaign**
Graduate fellowship support — Spring 2027

**Departmental Fellowship, University of Illinois Urbana-Champaign**
Graduate fellowship support — Fall 2025–Summer 2026

## Teaching Experience

**University of Illinois Urbana-Champaign**
Teaching Assistant: Calculus I (4 semesters), Calculus II (4 semesters), Linear Algebra (2 semesters) — 2021–present

**Brigham Young University**
Teaching Assistant: Complex Analysis (1 semester)

## Professional Service

**Organizer, UIUC Geometric Topology Seminar** — University of Illinois Urbana-Champaign
Created and organize an ongoing seminar for junior researchers in geometric topology and geometric group theory — 2025–present